

European Energy Analytics

Electricity Generation, Transition and Carbon Emissions

About this report: This report analyses electricity generation, demand, energy transition, carbon emissions and installed renewable capacity across 41 European markets using monthly energy data from 2015 to 2026.

Countries covered: The dataset includes 41 European markets, covering EU and non-EU countries. Comparative analyses use 38 countries with sufficient data for 2019–2025, while installed solar and wind capacity is analysed for 10 countries. Coverage varies by indicator and data availability.

Key Analytical Questions

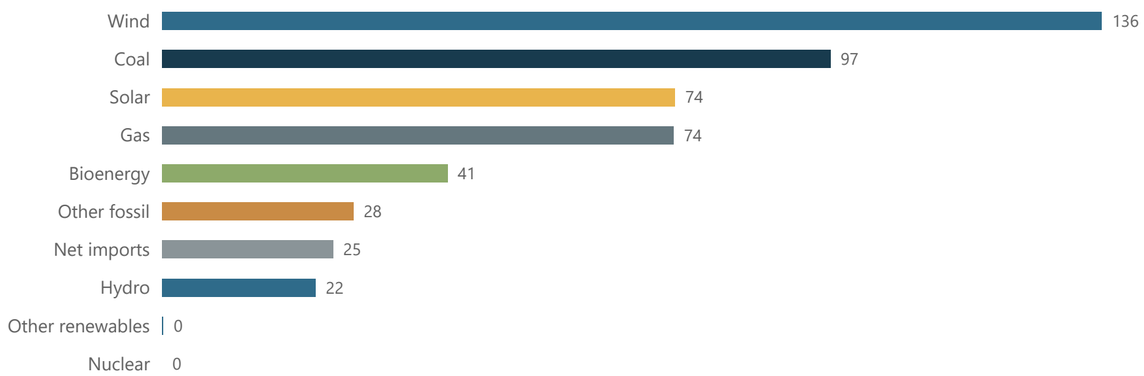
01. How have electricity generation and demand evolved across European countries?
02. How has Europe's electricity generation mix changed over time?
03. How have CO_2 emissions and carbon intensity developed over time?
04. How has installed solar and wind capacity evolved across European countries?

2024

Germany

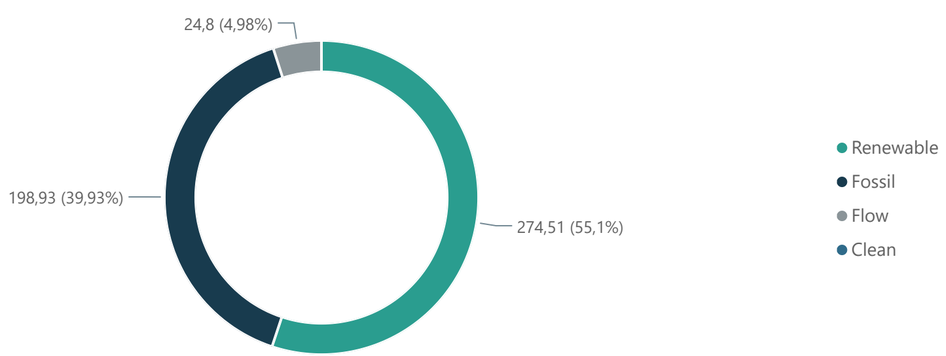
Power Generation (TWh)	Electricity Demand (TWh)	CO_2 Emissions ($MtCO_2$)	Carbon Intensity (gCO_2/kWh)	Installed Capacity (GW)
498,24	498,26	168,83	354,56	173,59

Electricity Generation Mix (TWh)

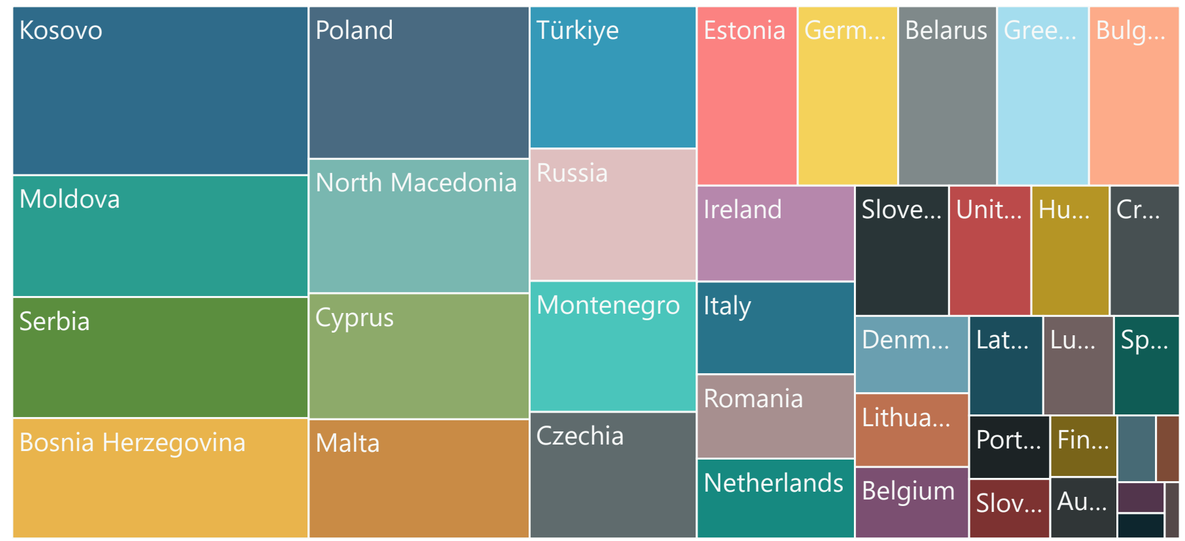


Generation and Total Demand (TWh)

Electricity Generation Mix by Category (TWh)

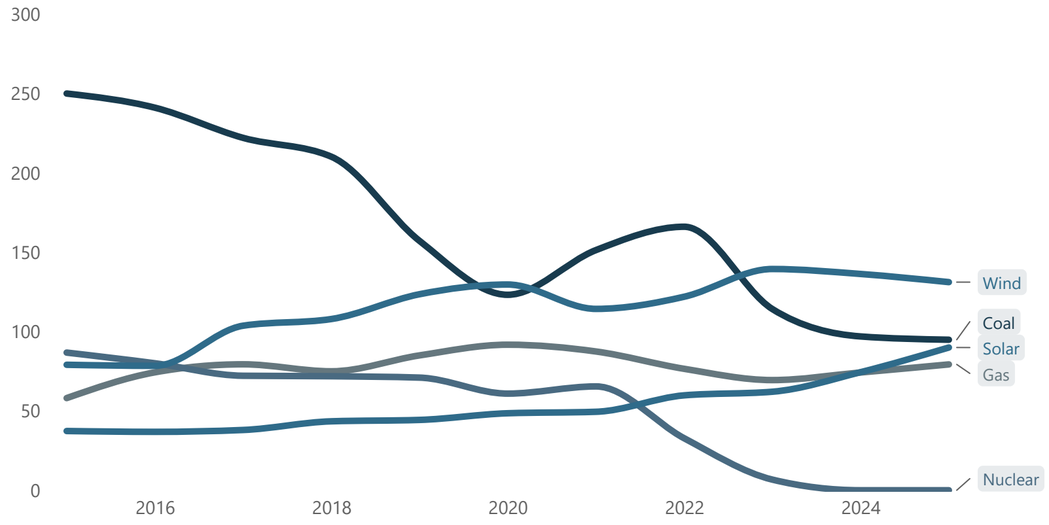


Carbon Intensity by Country (gCO_2/kWh)

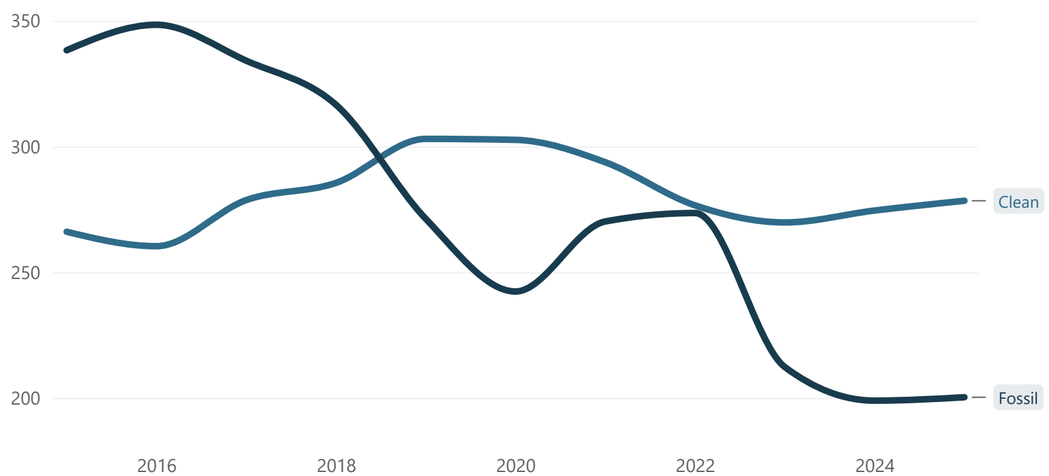


Germany

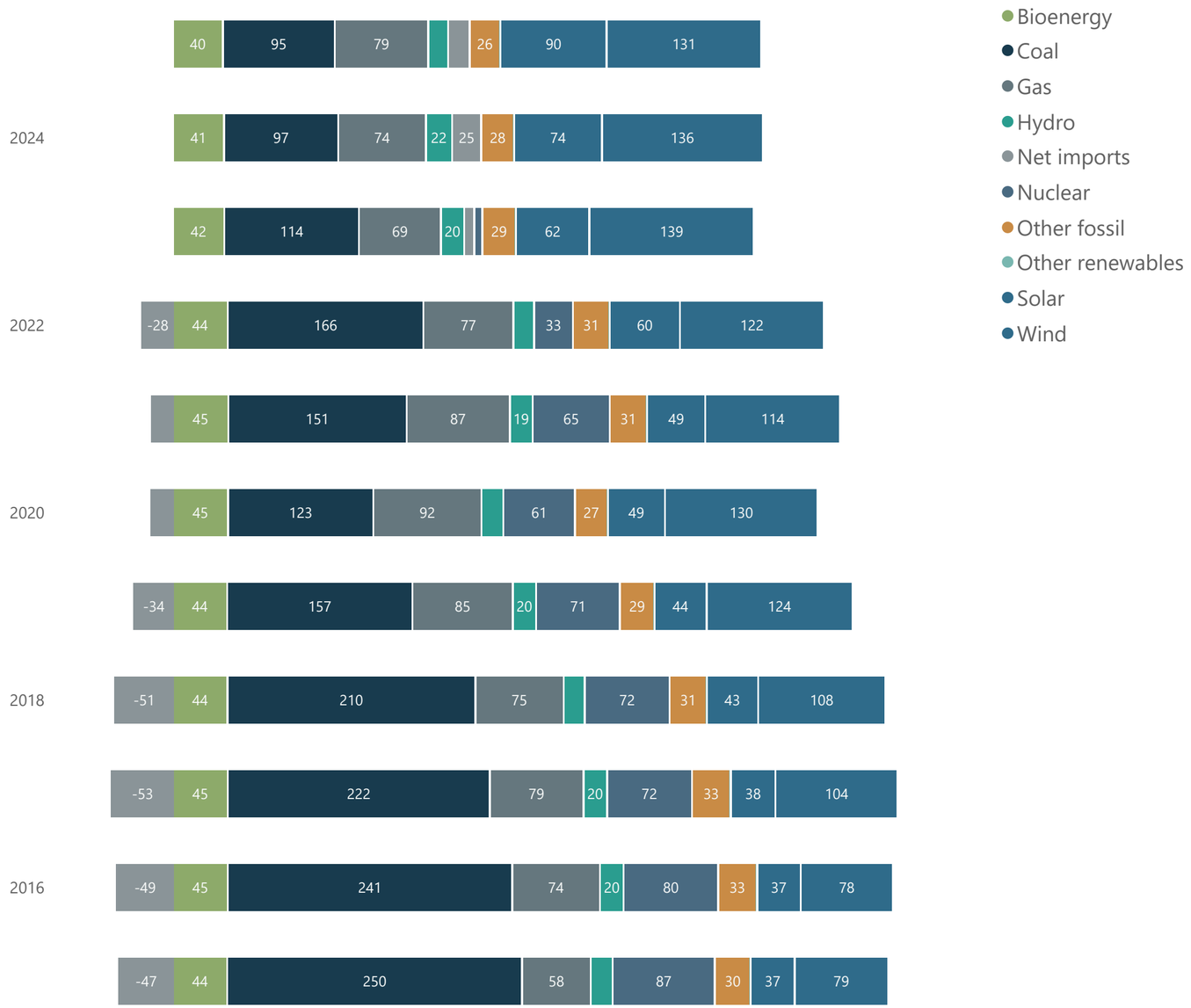
Generation Trends by Energy Source (TWh)



Clean vs Fossil Generation (TWh)



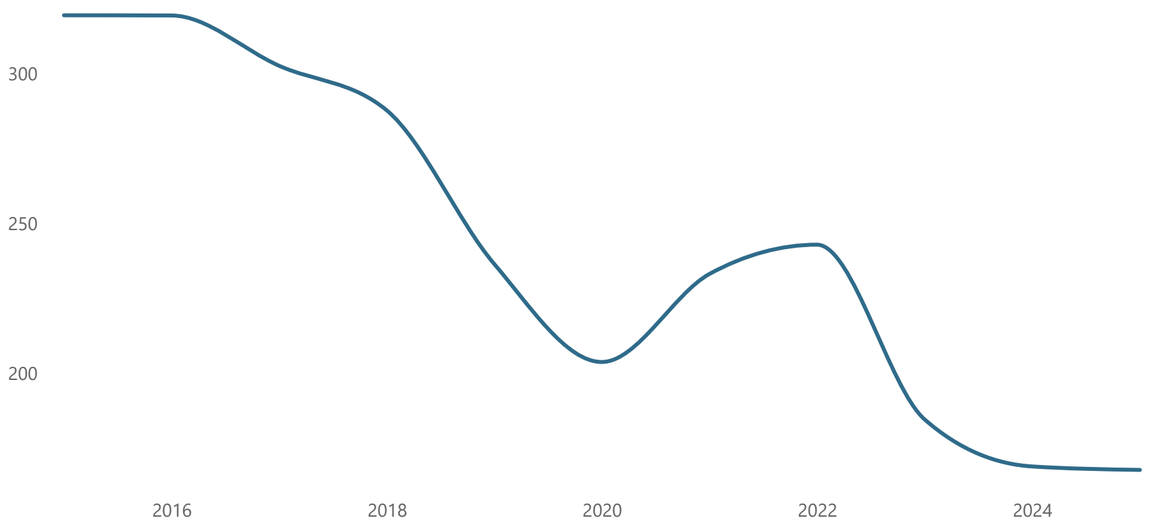
Electricity Generation Mix (TWh)



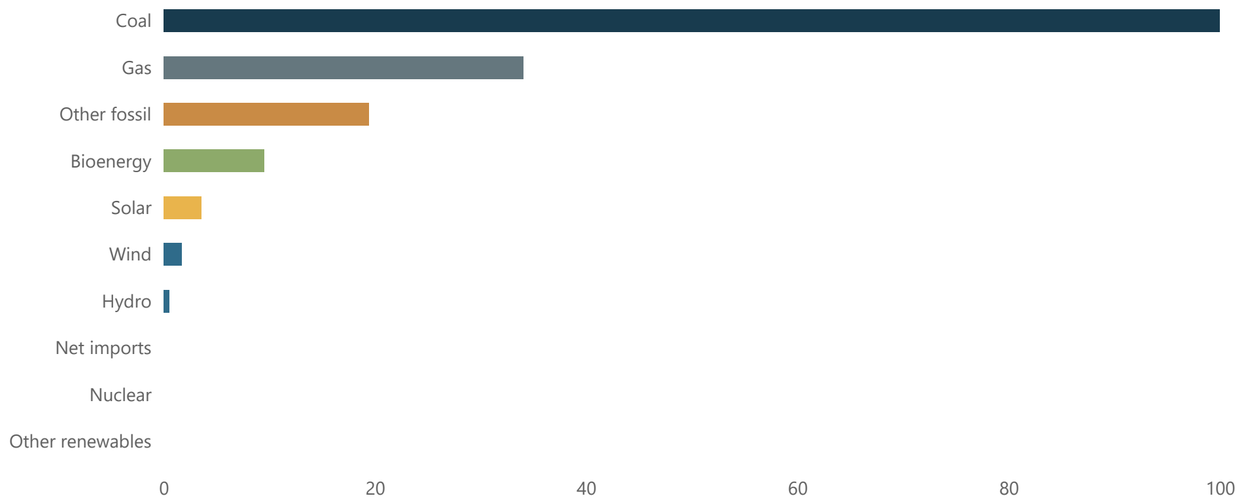
2024

Germany

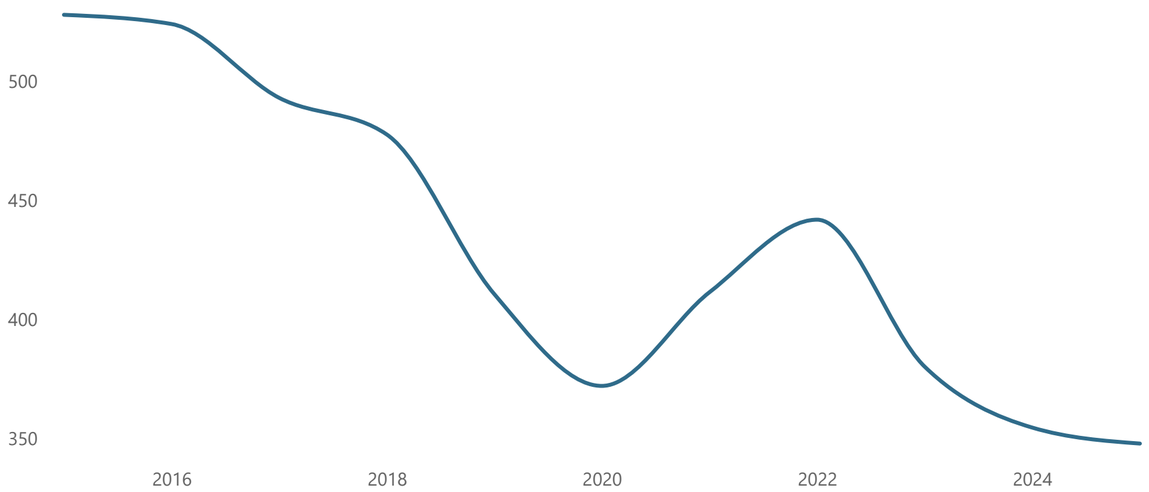
CO_2 Emissions Trend ($MtCO_2$)



CO_2 Emissions by Energy Source ($MtCO_2$)



Carbon Intensity Trend (gCO_2/kWh)



Carbon Intensity Change

-34,1 %

CO2 Emissions Change

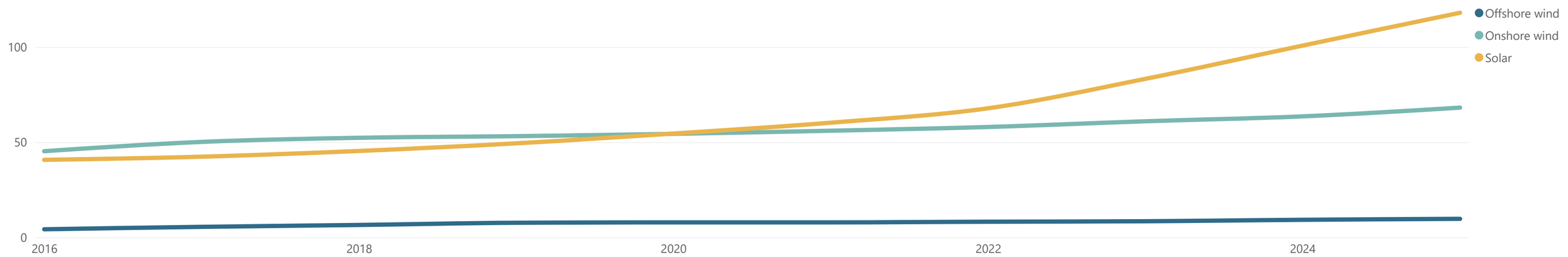
-47,5 %

2015 → 2025

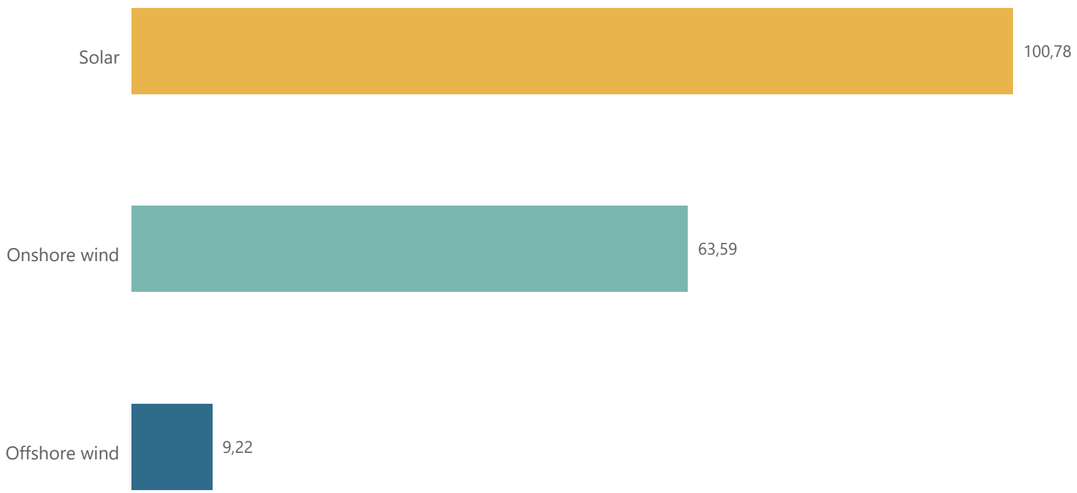
2024

Germany

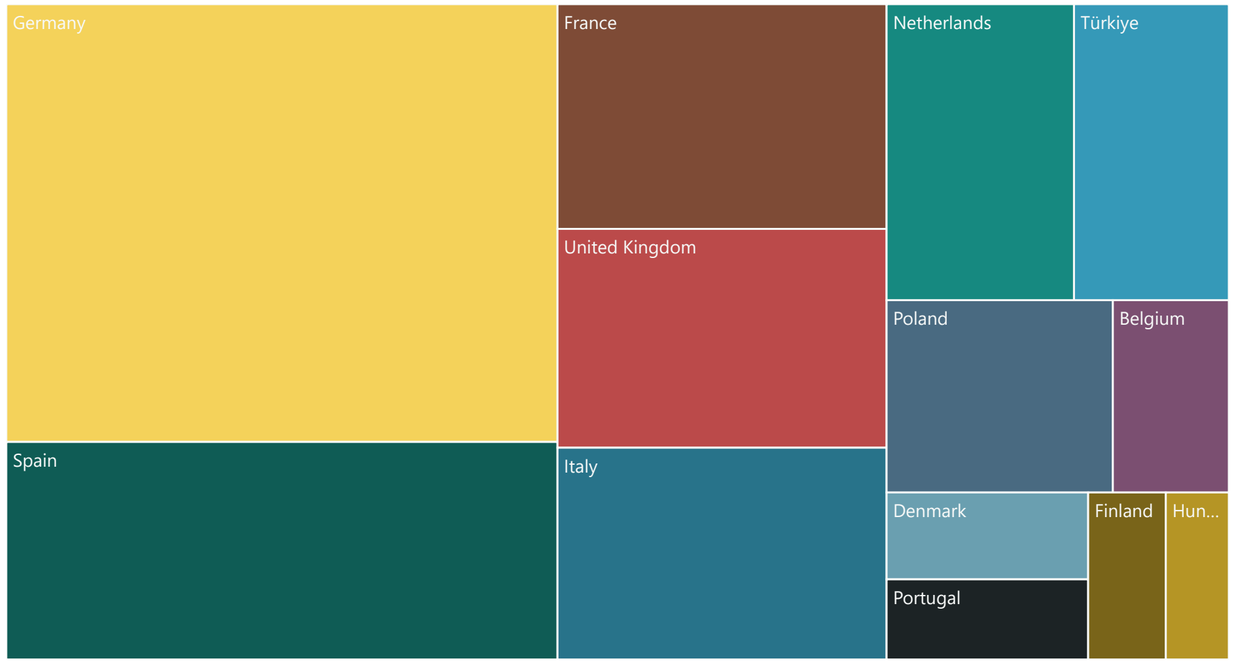
Installed Capacity Trend by Technology (GW)



Installed Capacity by Technology (GW)



Installed Capacity by Country (GW)



Key Insights & Conclusions

01 - Generation & Demand

Electricity generation and demand followed different development patterns across the 38 countries with comparable data between 2019 and 2025. Changes in domestic electricity production did not necessarily match changes in consumption, highlighting differences in national electricity systems. Net electricity imports and exports provide important additional context when interpreting these developments.

02 - Energy Mix & Transition

The electricity generation mix underwent significant changes between 2019 and 2025 across the 38 countries analysed. Renewable, fossil and other clean-energy shares evolved differently across national electricity systems, reflecting varying starting conditions and transition pathways. Changes are measured in percentage points to enable consistent comparisons between countries.

03 - Emissions & Carbon Intensity

Power-sector emissions and carbon intensity provide complementary perspectives on electricity-sector decarbonisation. The comparison of 38 countries between 2019 and 2025 reveals differences in national emissions trajectories and the carbon intensity of electricity generation. Evaluating both indicators helps distinguish changes in the total volume of emissions from changes in the emissions intensity of electricity production.

04 - Installed Capacity

Installed solar and wind capacity is compared across 10 countries with sufficient comparable data between 2019 and 2025. The analysis distinguishes between absolute capacity additions and relative growth, allowing countries with different initial capacity levels to be compared. Installed capacity measures the scale of generation infrastructure rather than actual electricity output.

01 – Generation & Demand: *Comparison based on the common complete period 2019–2025. Changes in net imports and exports are considered when interpreting differences between domestic generation and demand.*
02 – Energy Mix & Transition: *Comparison based on the common complete period 2019–2025. Energy-mix shares exclude net imports and aggregate series to avoid double counting.*
03 – Emissions & Carbon Intensity: *Comparison based on the common complete period 2019–2025. Total emissions refer to the aggregate "Total generation" emissions series; carbon intensity represents average monthly gCO_2/kWh.*
04 – Installed Capacity: *Comparison covers 2019–2025 and the ten countries with complete comparable Solar and Wind capacity data. Capacity values represent annual averages of monthly installed capacity.*